Pediatric Asthma Exacerbation

Section 1: Case Summary

Scenario Title:	Severe Pediatric Asthma Exacerbation
Keywords:	Pediatric, Asthma, Airway Management
Brief Description of Case:	A 5-year-old male with a history of asthma presents with a three-days of cough, wheeze and worsening shortness of breath. The team must recognize severe asthma and initiate usual asthma treatment, but the child does not respond to these basic treatments and continues to worsen. The team should escalate management – epinephrine, magnesium, ketamine. The patient continues to tire and requires intubation. Post-intubation, the team must optimize ventilator settings/paralyze/bear hug. If not treated aggressively, the patient will become hypotensive and increasingly hypoxic potentially leading to arrest.

Goals and Objectives	
Educational Goal:	To expose learners to severe asthma in the pediatric patient
Objectives: (Medical and CRM)	1. Displays leadership by maintaining calm demeanor during crisis and acting decisively 2. Employs good communication skills by using closed loop, listening to the input of others, and addressing concerned family members 3. Implements the basic ED treatment of asthma 4. Recognizes refractory/severe asthma and institutes appropriate treatments 5. Recognizes need for and demonstrates ability to intubate the pediatric asthma treatment 6. Manages post-intubation ventilation settings and appropriate sedation
EPAs Assessed:	

Learners, Setting and Personnel				
Target Learners:	☐ Junior Learners	☒Senior Learners	☐ Staff	
	☐ Physicians	☐ Nurses	☐ RTs	☐ Inter-professional
	☐ Other Learners:			
Location:	☒ Sim Lab	☐ In Situ	☐ Other:	
Recommended Number of Facilitators:	Instructors: 1			
	Sim Actors: 1			
	Sim Techs: 1			

Scenario Development	
Date of Development:	11/11/2017
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Last Revision Date:	
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Version Number:	

Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Charlie Jansen			Age: 5 years	Gender: M	Weight: 25 kg
Presenting complaint: Shortness of Breath					
Temp: 37.8	HR: 160/min	BP: 80/50	RR: 40/min	O_2Sat: 99% NRB	FiO_2:
Cap glucose: 6.2 mmol/L			GCS: 15		
Vignette to read aloud before the start of the case: A 5-year-old boy arrives via EMS with increased work of breathing. He has known asthma and has been using his puffer more over the past 3 days. He has been given 2.5mg of nebulized salbutamol on route with paramedics. Current vitals are: HR 160, BP 85/60, RR 40, O2 98 % on NRB, Temp 37.8 He has some ongoing wheeze noted by EMS. Triage note: Increasing wheeze and shortness of breath over the past 3 days. Worsening today not relieved by puffers. Single dose of nebulized salbutamol with EMS. Appears unwell.					
Allergies: none					
Past Medical History: Asthma – last steroids was last winter Eczema Hospitalized x 1 week @ 10 months with bronchiolitis			Current Medications: Flovent Ventolin		

Section 2B: Extra Patient Information

A. Further History	
Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, sim actor, SP, etc.)? Additional History to be provided by Father/Mother (sim actor): See below Social History: Lives with parents Family History: Mom and dad both have asthma and eczema ROS: HEENT: Rhinorrhea, other normal, Resp: Decreased AE, Audible wheeze	
B. Physical Exam	
Cardio: Normal HS, no murmurs	Neuro: Alert. Oriented. Appears scared.
Resp: audible wheeze. Chest tight. Tracheal tug. Intercostal indrawing. ++ increased work of breathing	Head & Neck: Nil.
Abdo: Nil.	MSK/skin: Nil.
Other: Nil.	

Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin *(Pediatric)*
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid

B. Special Equipment Required
Monitors: EKG Leads/Wires, NIBP Cuff, Pulse Oximeter, Temperature Probe, Defibrillator Pads Equipment: Gloves, Stethoscope, IV Bags/Lines, IV Push Medications, Nasal Prongs, Venturi Mask, Non-rebreather Mask, Bag Valve Mask, Laryngoscope, ET Tubes, Needle cric supplies, Peds NRB, PEEP Valve

C. Required Medications
Methylprednisolone, Normal Saline, Salbutamol, Ipratropium, Epinephrine, Magnesium Sulfate, Ketamine, Rocuronium

D. Moulage
None

E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed ☒ Patient not yet on monitor

F. Patient Reactions and Exam
Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format. Airway: Nil Breathing: Increased WOB, decreased air entry bilaterally, audible wheeze Circulation: Normal heart sounds, afebrile Disability: Moving all 4 limbs, GCS 15

Section 4: Sim Actors and Standardized Patients

Sim Actors and Standardized Patient Roles and Scripts	
Mother/Father	Mother or Father will come in shortly after the patient arrives. They will help answer any further history questions. HPI: He has known asthma and has been using his puffer more over the past 3 days. Past Medical History: asthma – last steroids was last winter, eczema, hospitalized x 1 week @ 10 months with bronchiolitis Current Medications: Ventolin, Flovent Allergies: none Social History: Lives with parents Family History: Mom and dad both have asthma and eczema ROS: HEENT: Rhinorrhea, other normal, Resp: today coughing, audible wheezing, with difficulty breathing As the Charlie gets more sick, the mother/father will remain at bedside and become more agitated unless they are consoled. Mother/father will notice Charlie getting sleepier and ask "Why does he look so tired" After intubation/ROSC mother/father should be updated by a team member. If not, she will become upset.

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: NSR HR: 160 bpm BP: 80/50 RR: 40/min O_2SAT: 99%NRB T: 37.8°C GCS: 15	Significant work of breathing. Mother concerned at bedside.	<u>Expected Learner Actions</u> ☐ Team inappropriate PPE ☐ Monitors, O2, IV access ☐ Methylprednisolone (1 –2mg/kg) ☐ IV NS bolus (20 cc/kg) ☐ Salbutamol (5 mg) and Ipratropium (0.5 mg) nebs x 3 ☐ IV MgSO4 (50 mg/kg) ☐ Cap glucose ☐ Portable CXR	<u>Modifiers</u> - NS bolus → HR 130 BP 85/55 - Mother not consoled → increasing agitation/ obstructive <u>Triggers</u> - Nebs/steroids/Mg → Next Phase - 5 minutes → Next Phase	Consider involving Social Work/Child Life Specialist depending on availability
2. Worse Dyspnea HR: 140 BP: 80/50 RR: 45 O_2SAT: 96% neb GCS: 13 (E3V4M6)	Patient complaining of worsening shortness of breath, more confused and unable to answer even simple questions	<u>Expected Learner Actions</u> ☐ IM epinephrine (0.01mg/kg/dose) ☐ Order labs, CXR, VBG ☐ Start IV epinephrine infusion at (0.1mcg/kg/min) ☐ Consider ketamine (2mg/kg bolus followed by 2mg/kg/hr) ☐ Salbutamol infusion ☐ Consider Heliox	<u>Modifiers</u> - VBG → 2 min later give result - Epinephrine → BP to 90/60 <u>Triggers</u> - Epi infusion started and MGSO4 given → fatigue - 5 minutes or none of the above interventions → fatigue	Nebulized MGSO4 (2.5ml) may be considered, though the evidence is weak.
3. Fatigue HR: 150 RR: 12 BP: 75/50 O_2SAT → 88% GCS: 10 (E2V3M5)	Patient becomes more unresponsive, confused, and tired appearing. Decreased resp effort. Mom asks why he looks so tired	<u>Expected Learner Actions</u> ☐ Prepare for intubation ☐ Call Anesthesia/Peds ICU (depending on leader) ☐ Push-dose Epi OR increase infusion rate ☐ Confirm tube placement ☐ Ventilation settings articulated for permissive hypercapnia	<u>Triggers</u> - Intubation → post-intubation hypotension -No intubation > 3 minutes → progressive hypoxia (from decreased resp rate) bradycardia, arrest	Consider whether to paralyze or not given the ventilation needed to counteract the acidosis Ventilation settings for permissive hypercapnia: Low TV and RR allowing for long I:E time

4. Post-intubation Hypotension HR → 130 O_2SAT → 90% RR → as per bagging rate BP →60/30 ****Vent Alarms****	Difficulty to bag	<u>Expected Learner Actions</u> ☐ Bag ventilation **(difficult to bag with poor chest rise)** ☐ Recognize post-intubation hypotension/hypoxia ☐ Disconnect from vent – bear hug ☐ Confirm tube ☐ Consider pneumonthorax/ obstructed tube ☐ Ensure paralysis ☐ Give Ventolin through ET tube	<u>Triggers</u> - All actions complete→ **6. Resolution** - Does not complete all actions within 5 minutes → **5. Bradycardia**	
5. Bradycardia O_2SAT → 60% over 1 min of bagging HR → 35 over 1 min		<u>Learner Actions</u> ☐ Starts CPR when HR < 30 ☐ Gives arrest dose epinephrine	<u>Triggers</u> - CPR without altering resp mechanics (2 minutes) – **End Case** **-** CPR with altering resp mechanics – HR and O2 improve, ROSC – **END Case**	
6. Resolution HR: 130/min BP: 90/50 RR: 8 (vented) O_2SAT: 95%		<u>Learner Actions</u> ☐ Post-intubation sedation ☐ ICU Consult, consideration ECMO/Sevofluorane ☐ Update mother	<u>Modifier</u> - Mother not updated → gets very upset. <u>Triggers</u> - ICU consult → **END CASE**	

Appendix A: Laboratory Results

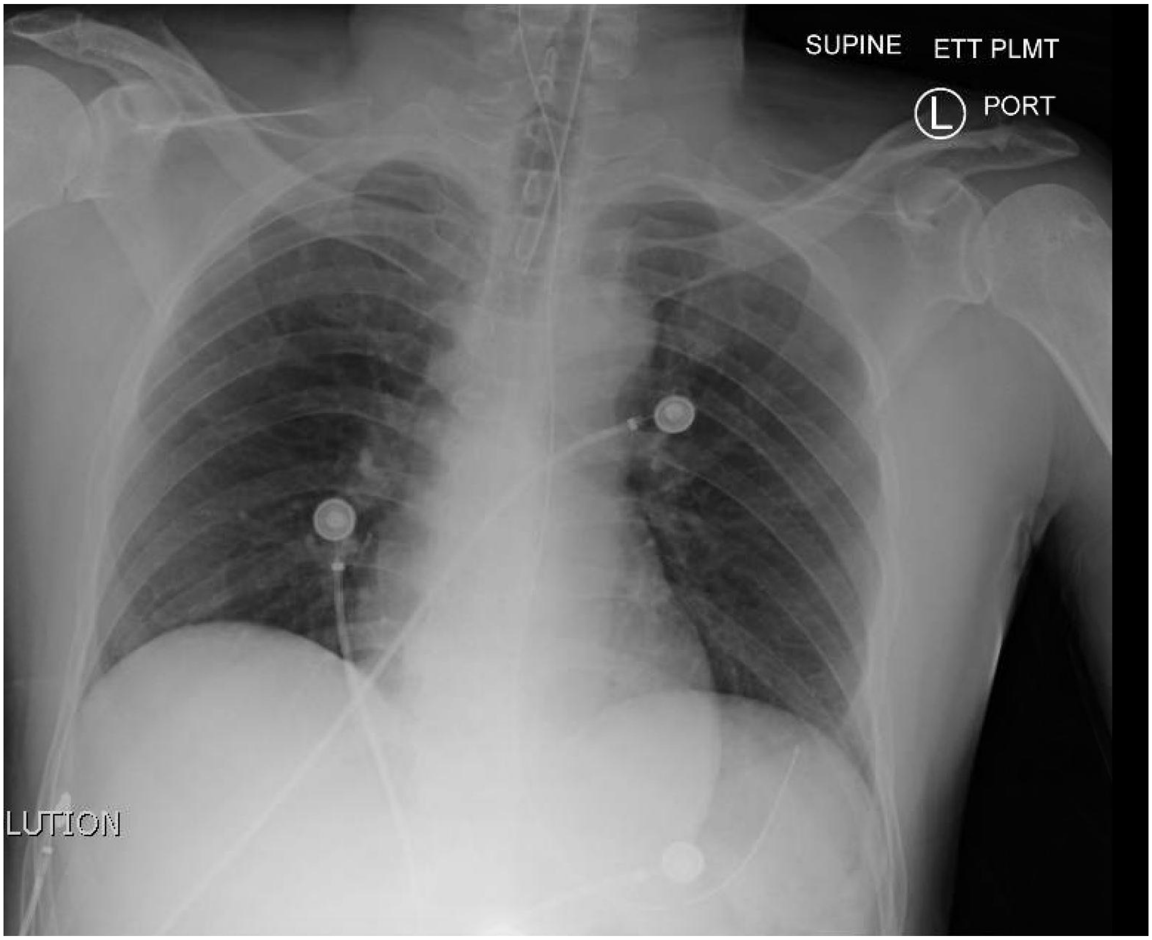
VBG	
pH	7.21
pCO_2	75
pO_2	55
HCO_3	20
Lactate	3.7

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Initial CXR: Normal Pediatriac Xray

CXR source: http://radiology-information.blogspot.ca/2015/04/normal-chest-x-ray.html

Post-intubation - normal



CXR source: http://jetem.org/ettcxr/

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

The debrief should be done as a group.

Sample Questions for Debriefing:

1. When did you decide to start secondary treatments for asthma? What are their indications? How do you determine whether this is a severe exacerbation?
2. What are the additional treatments (other than steroids and salbutamol) that are available for the treatment of severe asthma?
3. What are the indications for intubation in asthma?
4. What are your peri-intubation considerations?
5. What is your approach to post-intubation hypotension/decompensation?
6. What are the ventilation considerations after intubation in asthma patients?
7. Why did this patient arrest? (If he arrested in the scenario)
8. How did it feel as a team to manage this extremely unwell child that was not responding to the usual treatments that make most kids better? Do you feel you remained calm?

Key moments:

1. Recognition and treatment of severe asthma
2. Decision to intubate
3. Recognition of post-intubation complications

References

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